Notes version: 13 Nov 05:53 IST (if this time is old, reload the page)

GEOGRAPHY · India: climate and monsoon

UNDERSTAND IT FIRST

What is it? India has a **tropical monsoon climate**. The word **monsoon** (from Arabic *mausim*, 'season') means winds that **reverse direction with the seasons**: moist **south-west winds** from the sea in summer and dry **north-east winds** from the land in winter. Nearly **half of India's farmland is rain-fed**, so a good or poor monsoon affects **food prices, rural incomes, power and drinking water**. Learn it as **factors of India's climate → how the monsoon works → four seasons → onset, branches, breaks and retreat → rainfall pattern → importance → AP**.

1. Factors and mechanism

- **Factors of India's climate: latitude** (the Tropic of Cancer runs through the middle); the **Himalayas** (they **block cold Central Asian winds** in winter and **trap the monsoon**, making India warmer and wetter than other places at the same latitude); **distance from the sea** (coasts have a mild, **maritime** climate; the interior has **extreme, continental** temperatures); **altitude** (hill stations are cooler); **relief** (windward and rain-shadow sides); **pressure and winds**; **jet streams**; and ocean conditions (**El Niño, La Niña, Indian Ocean Dipole**).

What drives the south-west monsoon

2. The four seasons

India's four seasons

Season	Features
Cold weather (December–February)	Clear skies, cool dry **north-east winds**; **western disturbances** from the Mediterranean bring light rain to the north-west, which helps **rabi wheat**, and snow to the Himalayas; the **Tamil Nadu coast** gets rain from the north-east winds crossing the Bay of Bengal.
Hot weather (March–May)	Temperatures rise from south to north; **loo** winds in the north; pre-monsoon storms: **mango showers** (Kerala, Karnataka), **blossom showers** (Karnataka coffee), **Kal Baisakhi** (West Bengal) and **Bordoichila** (Assam).

South-west monsoon (June–September)	**Onset** over **Kerala around 1 June** (it came on **24 May in 2025**); covers the whole country by about **8 July**. Two branches: the **Arabian Sea branch** (stronger; heavy rain on the **Western Ghats**, Mumbai) and the **Bay of Bengal branch** (heavy rain in the **north-east**, Mawsynram; then moves west along the Himalayas). It gives about **three-quarters** of India's annual rain. The **long period average (LPA)** is about **87 cm**; 96–104% of LPA is 'normal'. Dry spells within the season are **breaks in the monsoon**. The **Tamil Nadu coast** stays fairly dry because it is in the **rain shadow** and the winds blow **parallel** to it.
Retreating monsoon (October–November, into December)	Withdrawal starts from **north-west India around mid-September** and is over by about **mid-October**. Clear skies with hot, humid days cause **'October heat'**. The **north-east monsoon** brings most of the year's rain to the **Tamil Nadu coast**, **south coastal AP** and **Rayalaseema**; **cyclones** from the Bay of Bengal are common.

3. Rainfall pattern and importance

- **Heavy rain (over 200 cm):** the **windward Western Ghats** and the **north-east** (Mawsynram, Cherrapunji). **Low rain (under 60 cm): western Rajasthan**, **Ladakh** and the **rain-shadow interior of the Deccan** (parts of Maharashtra, Karnataka and **Rayalaseema**). Rainfall is **uneven and unreliable**: the drier the area, the more it varies from year to year, which brings **droughts and floods**.
- **Importance:** the monsoon decides **kharif sowing**, refills **reservoirs and groundwater**, and affects **hydropower, inflation and rural demand**; it also gives India a **cultural unity** of seasons and festivals. **IMD** issues a long-range forecast in **April** and an update in **late May**; the **Monsoon Mission (2012)** and **Mission Mausam (2024)** aim to improve forecasts.
- **Andhra Pradesh** gets rain from **both monsoons**: the **south-west** monsoon gives the larger share to **north coastal AP and the Godavari–Krishna region**, while the **north-east** monsoon matters greatly for **Nellore, Prakasam and Rayalaseema** (Tirupati, Chittoor). Anantapur, in the rain shadow, is among India's **driest** districts.

EXAM TRAP

Factors: **latitude, Himalayas (block cold winds, trap monsoon), distance from sea, altitude, relief, winds, jet streams, El Niño, IOD**; mechanism: **land heats → low pressure; ITCZ/monsoon trough north; Tibet heating; westerly jet jumps north, easterly jet; SE trades cross equator → SW**; seasons: **cold (Dec–Feb; western disturbances, TN rain)**, **hot (Mar–May; loo, mango showers, blossom showers, Kal Baisakhi, Bordoichila)**, **SW monsoon (Jun–Sep; Kerala ~1 June, 24 May in 2025; whole country ~8 July; Arabian Sea branch stronger; Bay branch; ~75% of rain; LPA ~87 cm; breaks; TN dry)**, **retreating (Oct–Nov; withdrawal from mid-Sept; 'October heat'; NE monsoon on TN, south AP, Rayalaseema; cyclones)**; heavy **>200 cm** (Western Ghats, NE); low **<60 cm** (W Rajasthan, Ladakh, Deccan interior, Rayalaseema); IMD **April and late May** forecasts; **Monsoon Mission 2012, Mission Mausam 2024**; AP: both monsoons; **Anantapur** very dry.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. The Arabian Sea branch of the south-west monsoon is stronger than the Bay of Bengal branch.

2. The south-west monsoon brings about three-quarters of India's annual rainfall.

3. The Tamil Nadu coast gets most of its rain during the south-west monsoon.

4. Western disturbances bring winter rain to north-west India.

Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. The Tamil Nadu coast gets most of its rain from the north-east (retreating) monsoon in October–December.

PRACTICE

Match List-I (pre-monsoon shower) with List-II (region):

List-I (pre-monsoon shower)	List-II (region)
A. Mango showers	1. Assam
B. Blossom showers	2. West Bengal
C. Kal Baisakhi	3. Coffee areas of Karnataka
D. Bordoichila	4. Coastal Kerala and Karnataka

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 1 D → 2

(4) A → 4 B → 3 C → 2 D → 1

Answer: (4) Mango showers: Kerala–Karnataka coast; blossom showers: coffee; Kal Baisakhi: Bengal; Bordoichila: Assam.

PRACTICE

Which one of the following statements is NOT correct?

A. The Himalayas let cold Central Asian winds into India in winter.

B. The south-west monsoon normally reaches Kerala around 1 June.

C. Mawsynram is among the wettest places on Earth.

D. Shillong lies in a rain-shadow area.

Answer: A. The Himalayas block the cold Central Asian winds, keeping north India warmer in winter.

PRACTICE

Assertion (A): The retreating-monsoon season is known for 'October heat' in many parts of India.
Reason (R): In October the sky over most of India is overcast with continuous heavy rain.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true. R is false: clear skies, high temperature and humid ground cause the oppressive 'October heat'.

PRACTICE

A severe cyclonic storm crosses the Nellore coast in mid-November, bringing heavy rain to Nellore and Tirupati districts. In which season does this happen?

A. South-west monsoon

B. Retreating (north-east) monsoon

C. Hot weather season

D. Cold weather season

Answer: B. October–December is the retreating/north-east monsoon season, when Bay of Bengal cyclones hit south AP and Tamil Nadu.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the factors of India's climate and the five drivers of the monsoon. 2. Write the four seasons with their features, including onset, branches, LPA and withdrawal. 3. Write the rainfall zones, the importance of the monsoon and AP's pattern.

ECONOMY · Decentralisation of financial resources

UNDERSTAND IT FIRST

What is it? Fiscal federalism is about **who collects taxes and who spends**. The **Union** has the most productive taxes (income tax, corporation tax, a large share of GST), while **States and local bodies** run most services (police, health, schools, water, roads). This mismatch is the **vertical imbalance**; differences between rich and poor States are the **horizontal imbalance**. The Constitution fixes it by **sharing taxes and giving grants**, mainly through the **Finance Commission**, the **GST system** and, within States, the **State Finance Commission**. Learn it as **the imbalance → constitutional provisions → Finance Commissions → GST → State and local finances → AP**.

How money flows down to local bodies

1. Constitutional provisions (Part XII)

Article	Provision
268, 269	**268**: duties **levied by the Union** but **collected and kept by States** (e.g. some stamp duties). **269**: taxes levied and collected by the Union but **assigned to States** (e.g. on inter-State sale of goods outside GST).
269A, 279A	**269A**: **IGST** on inter-State supply, shared between the Union and States. **279A**: the **GST Council**: chaired by the **Union Finance Minister**; the Centre has **one-third** and the States **two-thirds** of the vote; decisions need a **three-fourths majority**.
270, 271	**270**: taxes levied and collected by the Union are **shared with States** (the **divisible pool**), as the Finance Commission recommends. **271**: **surcharges** for Union purposes are **not shared**; **cesses** are also outside the divisible pool, which States criticise.
275, 282	**275**: **statutory grants-in-aid** to States in need (e.g. **revenue deficit grants**) on Finance Commission advice. **282**: **discretionary grants** by the Union for **any public purpose** (used for **centrally sponsored schemes**).
276, 293	**276**: States and local bodies may levy a **professional tax** up to **₹2,500 a year**. **293**: a State that **owes money to the Centre** needs the **Centre's consent to borrow**.
280; 243I, 243Y	**280**: the **Finance Commission**, set up by the **President every five years** (a chairman and four members). After the 73rd and 74th Amendments, it also suggests measures to **augment State funds**

	for panchayats and municipalities (Art 280(3)(bb) and (c)), based on **State Finance Commission** reports (**Arts 243I, 243Y**).

2. Finance Commissions

- The **1st Finance Commission (1951)** was chaired by **K.C. Neogy**. The **14th (Y.V. Reddy; 2015–20)** raised the States' share of the divisible pool sharply to **42%**. The **15th (N.K. Singh; 2021–26)** gave **41%** (1% less because **J&K** became a UT), and about **₹4.36 lakh crore** in grants to **local governments**, part of them **tied** to **sanitation and drinking water**. The **16th (Arvind Panagariya; 2026–31)** was set up in **December 2023** (check its recommended share in the latest Budget).
- **Horizontal sharing** among States uses a **formula**. Under the **15th FC**, **income distance** (poorer States get more) had the biggest weight, with **population (2011)**, **area**, **forest and ecology**, **demographic performance** (rewarding States that controlled population) and **tax effort**. **Andhra Pradesh's share** of the States' pool under the 15th FC was about **4.05%**; AP also received **revenue-deficit grants** after bifurcation.

15th Finance Commission criteria

3. GST, State and local finances

- **GST** came through the **101st Amendment (2016)** and started on **1 July 2017**, merging many Union and State taxes. It has **CGST** (Union), **SGST** (State) and **IGST** (inter-State). States were promised **compensation for revenue loss for five years** (till June 2022); the **compensation cess** was continued to repay loans taken during COVID. In **September 2025**, GST was simplified to mainly **two rates (5% and 18%)** plus a higher rate for luxury and 'sin' goods (check the latest).
- **States' own revenue: SGST**, **excise on alcohol**, **stamp duty and registration**, **motor vehicle tax**, **electricity duty**, land revenue; States also **borrow** within **FRBM** limits (fiscal deficit target about **3% of GSDP**). **Local bodies' own revenue: property tax**, **professional tax**, **user charges and fees**, advertisement tax. Local bodies remain **weak**: own revenue is small, many States **delay SFCs** or ignore their reports, and most money comes as **tied grants**. The **15th FC** made grants conditional on **audited accounts** and **property tax floor rates**.

EXAM TRAP

Vertical imbalance (Union collects more, States spend more); **horizontal** (rich vs poor States); **268** (levied by Union, collected by States), **269** (assigned), **269A (IGST)**, **270 (divisible pool)**, **271 (surcharges not shared)**; cesses outside pool; **275 statutory grants**, **282 discretionary grants**; **276 professional tax ≤ ₹2,500**; **293 borrowing consent**; **280 FC every 5 years (chair + 4)**; **280(3)(bb), (c)** local bodies; **SFC 243I, 243Y**; **1st FC K.C. Neogy 1951**; **14th Y.V. Reddy 42%**; **15th N.K. Singh 41% (2021–26)**, local grants **~₹4.36 lakh crore**; **16th Arvind Panagariya (2026–31)**; criteria: **income distance 45, population 15, area 15, forest 10, demographic performance 12.5, tax effort 2.5**; **AP ~4.05%**; **GST: 101st (2016), 1 July 2017, CGST/SGST/IGST, Council 279A (FM chair; Centre 1/3, States 2/3; 3/4 majority), compensation till June 2022, 2025: 5% and 18%**; State taxes (SGST, liquor excise, stamps, vehicle tax); FRBM **3%**; local taxes (property, professional, fees); weak local finances.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. The Finance Commission is constituted every five years under Article 280.

2. The 15th Finance Commission recommended 41% of the divisible pool for the States.

3. Cesses and surcharges are part of the divisible pool shared with States.

4. State Finance Commissions are provided for under Article 243I.

Which of the above are correct?

A. 1, 2 and 4 only

B. 1 and 2 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. Cesses and surcharges are kept outside the divisible pool, so States do not get a share of them.

PRACTICE

Match List-I (Article) with List-II (subject):

List-I (Article)	List-II (subject)
A. Article 270	1. Discretionary grants for any public purpose
B. Article 275	2. GST Council
C. Article 279A	3. Statutory grants-in-aid to States
D. Article 282	4. Sharing of Union taxes with States

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 2 C → 3 D → 1

(3) A → 4 B → 3 C → 2 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (3) 270: tax sharing; 275: statutory grants; 279A: GST Council; 282: discretionary grants.

PRACTICE

Which one of the following statements is NOT correct?

A. N.K. Singh chaired the 15th Finance Commission.

B. K.C. Neogy chaired the first Finance Commission.

C. The 14th Finance Commission raised the States' share to 42%.

D. Decisions of the GST Council must be unanimous.

Answer: D. GST Council decisions need a three-fourths majority of weighted votes, not unanimity.

PRACTICE

Assertion (A): Under Article 276, a professional tax cannot exceed ₹2,500 per person a year.
Reason (R): Property tax is the most important own source of revenue for municipalities.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: B. Both are true, but the limit on professional tax is not explained by the importance of property tax.

PRACTICE

A municipality and the panchayats in a district want a larger share of the State Government's own tax revenue. Which body recommends how State taxes should be shared with local bodies?

A. The Union Finance Commission

B. NITI Aayog

C. The State Finance Commission

D. The GST Council

Answer: C. Under Arts 243I and 243Y, the State Finance Commission recommends sharing State taxes and grants with panchayats and municipalities.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the two imbalances and the Article-wise table. 2. Write the Finance Commissions with chairs, shares and criteria, and AP's share. 3. Write GST facts and the State and local revenue sources and problems.

SCIENCE · Transgenics, eco-friendly agriculture and bioremediation

UNDERSTAND IT FIRST

What is it? Biotechnology uses living things to make useful products. A **transgenic** (genetically modified, **GM**) organism carries a **gene from another species**, put in by **recombinant DNA technology**, for example cotton with a bacterial gene that kills pests. **Eco-friendly agriculture** grows food while protecting soil, water and health, using **biofertilisers, biopesticides and natural farming** instead of heavy chemicals; **Andhra Pradesh leads India** in natural farming. **Bioremediation** uses **microbes and plants to clean up pollution**. Learn it as **how GM organisms are made → examples and India's status → regulation and concerns → eco-friendly farming → bioremediation**.

1. Transgenics

Making a transgenic organism

Item	Facts
Tools	**Restriction enzymes** ('molecular scissors'), **DNA ligase** ('molecular glue'), **vectors** such as **plasmids**; **Agrobacterium tumefaciens** is called a **'natural genetic engineer'** because its **Ti plasmid** carries DNA into plant cells; the **gene gun** shoots DNA-coated particles into cells. **Gene editing** with **CRISPR-Cas9** (Nobel Prize **2020**, **Charpentier and Doudna**) changes a plant's own genes without adding foreign DNA; India **exempted** such simple edits (**SDN-1, SDN-2**) from GM rules in **2022** and released two **genome-edited rice varieties** in **May 2025**.
GM crops	**Bt cotton** (**cry** genes from **Bacillus thuringiensis** kill **bollworms**; approved in **2002**) is the **only GM crop commercially grown in India**; **pink bollworm** later developed resistance. **Bt brinjal** was cleared by GEAC in **2009** but put under a **moratorium in 2010** (Bangladesh grows it). **GM mustard (DMH-11)** got approval for **environmental release in 2022**; the **Supreme Court gave a split verdict (2024)** and asked for a **national GM policy**. **Golden rice** has **beta-carotene** (vitamin A); the first GM food was the **Flavr Savr tomato (1994)**.
Animals and medicine	**Human insulin** made by GM **E. coli** (**Humulin, 1982**, the first rDNA medicine); vaccines such as **hepatitis B**; **transgenic animals**: **Rosie** the cow (milk with a human protein), **GM salmon**, **GM mosquitoes** to fight dengue and malaria.

Regulation and concerns	The **Genetic Engineering Appraisal Committee (GEAC)** under the **Environment Ministry** (Rules of **1989** under the EPA) approves field trials and release; the **RCGM** under the **Department of Biotechnology** reviews research; internationally, the **Cartagena Protocol** on biosafety. **Concerns: gene flow** to wild relatives, pests becoming **resistant**, effects on **non-target insects**, **farmers' dependence** on seed companies, and **labelling** of GM food. **Benefits:** higher yields, less pesticide, better nutrition, drought tolerance.

2. Eco-friendly agriculture

- **Biofertilisers: Rhizobium** (nitrogen fixation in **legume** root nodules), **Azotobacter** and **Azospirillum** (free-living nitrogen fixers), **blue-green algae** and **Azolla** (a water fern hosting **Anabaena**) for **paddy**, and **mycorrhiza (VAM)** fungi that help roots absorb **phosphorus**. **Biopesticides: Bt**, the fungus **Trichoderma** (controls soil-borne plant diseases), **NPV** (a virus against caterpillars) and **neem**. **Integrated Pest Management (IPM)** combines resistant varieties, crop rotation, natural enemies and only need-based chemicals.
- **Organic and natural farming: organic farming** avoids synthetic fertilisers and pesticides; **Sikkim** became India's **first fully organic State (2016)**; schemes: **Paramparagat Krishi Vikas Yojana (PKVY, 2015)** for organic clusters and **MOVCDNER** for the north-east. **Natural farming** (inspired by **Subhash Palekar's** 'zero budget' method) uses on-farm inputs such as **jeevamrutha** (from cow dung and urine), **beejamrutha** (seed treatment), **mulching** and soil aeration. **Andhra Pradesh Community-managed Natural Farming (APCNF)**, run by **Rythu Sadhikara Samstha** since **2016**, is one of the world's largest agroecology programmes and won the **Gulbenkian Prize for Humanity (2024)**. The Centre launched the **National Mission on Natural Farming (November 2024)**.
- **Other steps: Soil Health Cards (2015)**; **neem-coated urea (2015)**; **nano urea (2021)**; **PM-PRANAM (2023)** rewards States that cut chemical fertiliser use; **micro-irrigation** under **PMKSY ('per drop more crop')**; the **System of Rice Intensification (SRI)**; zero tillage and crop diversification.

3. Bioremediation

Types of bioremediation

- **Bioremediation** uses **microbes, plants or fungi** to break down or remove pollutants such as **oil, pesticides, heavy metals and sewage**. It is cheap and natural but **slow**, and works only for some pollutants. **In-situ** methods treat the soil or water **where it is** (adding nutrients or microbes, pumping air); **ex-situ** methods move it to a treatment site (**landfarming, biopiles, bioreactors**).
- **Examples: Ananda Mohan Chakrabarty** created an oil-eating **Pseudomonas** ('superbug'); its **US patent (1980)** was the first patent on a **GM organism**. **'Oilzapper'** (developed by **TERI** with **Indian Oil**) is a mix of bacteria used to clean **oil spills and oily sludge**. **Phytoremediation: Indian mustard** takes up **heavy metals**; **sunflowers** were used to absorb **radioactive** elements after **Chernobyl**; **water hyacinth** and **constructed wetlands** clean sewage. **Biomining** of old dumps (Day 29) also uses microbes.

EXAM TRAP

rDNA: restriction enzymes (scissors), **ligase** (glue), vectors (plasmids); **Agrobacterium Ti plasmid ('natural genetic engineer')**; gene gun; **CRISPR-Cas9 (Nobel 2020)**; SDN-1, 2 exempt (**2022**); **genome-edited rice (May 2025)**; **Bt cotton 2002 (only commercial GM crop; Bacillus thuringiensis, cry; pink bollworm resistance)**; **Bt brinjal: GEAC 2009, moratorium 2010**; **GM mustard DMH-11 (2022; SC split verdict 2024)**; **golden rice (beta-carotene)**; **Flavr Savr 1994**; **Humulin 1982**; Rosie, GM salmon, GM mosquitoes; **GEAC (MoEFCC, Rules 1989)**, **RCGM (DBT)**, **Cartagena**; concerns; biofertilisers **Rhizobium, Azotobacter, Azospirillum, BGA, Azolla–Anabaena, mycorrhiza (P)**; biopesticides **Bt, Trichoderma, NPV, neem**; **IPM**; **Sikkim organic 2016**; **PKVY 2015**, MOVCDNER; natural farming (**Palekar**; jeevamrutha, beejamrutha); **APCNF 2016 (Rythu Sadhikara Samstha; Gulbenkian Prize 2024)**; **NMNF Nov 2024**; **Soil Health Card 2015, neem-coated urea 2015, nano urea 2021, PM-PRANAM 2023, PMKSY, SRI**; bioremediation **in-situ vs ex-situ**; **Chakrabarty superbug (patent 1980)**; **Oilzapper (TERI–IOC)**; **Indian mustard (metals), sunflower (Chernobyl)**, constructed wetlands.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:

1. Bt cotton is the only GM crop approved for commercial cultivation in India.

2. The Bt gene comes from the bacterium Bacillus thuringiensis.

3. The Genetic Engineering Appraisal Committee works under the Ministry of Environment, Forest and Climate Change.

4. Bt brinjal is widely grown by farmers across India.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2, 3 and 4

D. 1, 2 and 3 only

Answer: D. Bt brinjal has been under a moratorium in India since 2010.

PRACTICE

Match List-I (organism) with List-II (use in farming):

List-I (organism)	List-II (use in farming)
A. Rhizobium	1. Biological control of fungal plant diseases
B. Azolla	2. Helps roots absorb phosphorus
C. Mycorrhiza	3. Water fern used in paddy that hosts nitrogen-fixing Anabaena

D. Trichoderma	**4.** Nitrogen fixation in legume root nodules

(1) A → 3 B → 4 C → 2 D → 1

(2) A → 4 B → 3 C → 2 D → 1

(3) A → 4 B → 2 C → 3 D → 1

(4) A → 4 B → 3 C → 1 D → 2

Answer: (2) Rhizobium: legumes; Azolla: paddy; mycorrhiza: phosphorus; Trichoderma: biocontrol.

PRACTICE

Which one of the following statements is NOT correct?

A. Sikkim was declared India's first fully organic State.

B. APCNF is a natural farming programme of Andhra Pradesh.

C. Golden rice is enriched with iron.

D. Indian mustard can be used to remove heavy metals from soil.

Answer: C. Golden rice is enriched with beta-carotene (provitamin A), which gives it its golden colour.

PRACTICE

Assertion (A): Agrobacterium tumefaciens is widely used to make transgenic plants.
Reason (R): Its Ti plasmid can transfer a piece of DNA into the genome of plant cells.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Scientists replace the tumour genes of the Ti plasmid with useful genes, so it carries them into plants.

PRACTICE

After an oil spill near a port, scientists spread a mixture of bacteria that break down the oil into harmless substances. What is this method called?

A. Phytoremediation

B. Bioremediation using oil-degrading bacteria

C. Biomagnification

D. Incineration

Answer: B. This is bioremediation (bioaugmentation), like India's Oilzapper; phytoremediation uses plants.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the rDNA steps and tools, gene editing, and the GM crops, animals and medicines with India's status. 2. Write GEAC, RCGM, Cartagena and the concerns and benefits. 3. Write biofertilisers, biopesticides, IPM, organic and natural farming (APCNF) with schemes, and bioremediation types and examples.

NOTEBOOK PAGE · Day 32: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

GEOGRAPHY: Climate and monsoon

Area	Write
Controls and mechanism	Factors of India's climate; five drivers of the monsoon
Seasons and rain	Four IMD seasons with features; onset, branches, breaks, withdrawal dates, LPA; NE monsoon; rainfall zones; importance; AP pattern

ECONOMY: Financial decentralisation

- Vertical and horizontal imbalance; Articles 268–282, 293, 276; Finance Commissions (1st, 14th, 15th, 16th) with shares and criteria; GST (101st, Council, compensation, 2025 reform); State and local own revenues; SFCs and 280(3)(bb)/(c); local body grants; AP's share.

SCIENCE: Transgenics, eco-farming, bioremediation

Write: rDNA steps and tools; GM examples (crops, animals, medicines) with India's status; gene editing; GEAC, RCGM, Cartagena; concerns; organic and natural farming (APCNF, Sikkim, NMNF), biofertilisers, biopesticides, IPM, schemes; bioremediation types with examples (Chakrabarty, Oilzapper, phytoremediation).